

CHAPTER - 7

TEST OF HYPOTHESIS

7.1 INTRODUCTION

In the previous chapter we have seen how a parameter can be estimated from sample data. We can find either a single number for the parameter (a point estimate) or an interval of values (an interval estimate). However there are many problems, in which, rather than estimating the value of a parameter we need to decide whether to accept or reject a statement about the parameter. This statement is called a *hypothesis* and the decision-making procedure about the hypothesis is called *hypothesis testing*. This is one of the most useful aspects of statistical inference, since many types of decision-making problems, tests or experiments in the engineering world can be formulated as hypothesis - testing problems.

7.2 TEST OF HYPOTHESIS

The main object of the Sampling Theory is the study of the Tests of Hypothesis or Tests of Significance.

In many circumstances, we are to make decisions about population on the basis of only sample information. For example, on the basis of sample data,

- (i) a drug chemist is to decide whether a new drug is really effective in curing a disease,
- (ii) a quality control manager is to determine whether a process is working properly,
- (iii) a statistician has to decide whether a given coin is biased, etc.

Such decisions are called statistical decisions (or simply decisions)

Definition : Statistical Hypothesis

[JNTU (K) Dec. 2013 (Set No. 3)]

In many circumstances, to arrive at decisions about the population on the basis of sample information, we make assumptions (or guesses) about the population parameters involved. Such an assumption (or statement) is called a *Statistical Hypothesis* which may or may not be true. The procedure which enables us to decide on the basis of sample results whether a hypothesis is true or not, is called Test of Hypothesis or Test of Significance.

Since we use probability distributions to represent population, a statistical hypothesis may also be thought of as a statement about the probability distribution of a random variable. The hypothesis will usually involve one or more parameters of this distribution.

Example : 1. The majority of men in the city are smokers.

2. The teaching methods in both the schools are effective.

7.3 PROCEDURE FOR TESTING A HYPOTHESIS [JNTU NOV. 2008 (SET NO. 2)]

Test of Hypothesis involves the following steps :

Step 1 : Statement (or assumption) of hypothesis

There are two types of hypothesis :

- (i) Null Hypothesis
- (ii) Alternative Hypothesis.

(i) **Null Hypothesis :** For applying the tests of significance, we first set up a hypothesis - a definite statement about the population parameter. Such a hypothesis is usually a hypothesis of no-difference, is called Null Hypothesis.

It is in the form $H_0 : \mu = \mu_0$

μ_0 is the value which is assumed or claimed for the population characteristic. It is the reference point against which the Alternative Hypothesis is set up, as explained in the next step.

Definition : A null hypothesis is the hypothesis which asserts that there is no significant difference between the statistic and the population parameter and whatever observed difference is there, is merely due to fluctuations in sampling from the same population. It is always denoted by H_0 . To test whether one procedure is better than another, we assume that there is no difference between the procedures. Similarly to test whether there is a relationship between two variates, we take H_0 that there is no relationship.

For example, in case of a single statistic, H_0 will be that the sample statistic does not differ significantly from the hypothetical parameter value and in the case of two statistics (H_0) will be that the sample statistics do not differ significantly.

(ii) **Alternative Hypothesis :** Any hypothesis which contradicts the Null Hypothesis is called an *Alternative Hypothesis*, usually denoted by H_1 . The two hypothesis H_0 and H_1 are such that if one is true, the other is false and vice versa. For example, if we want to test the null hypothesis that the population has a specified mean μ_0 (say) i.e., $H_0 : \mu = \mu_0$, then the Alternative Hypothesis would be

(i) $H_1 : \mu \neq \mu_0$ (i.e., either $\mu > \mu_0$ or $\mu < \mu_0$)

or (ii) $H_1 : \mu > \mu_0$

or (iii) $H_1 : \mu < \mu_0$

The Alternative Hypothesis (i) is known as a two-tailed alternative and the Alternative Hypothesis in (ii) is known as right-tailed and in (iii) is known as left-tailed.

The setting of alternative hypothesis is very important to decide whether we have to use a single-tailed (right or left) or two-tailed test.

Alternate Hypothesis is in one of the following forms.

$$H_1 : \mu \neq \mu_0$$

$$\text{or } H_1 : \mu > \mu_0$$

$$\text{or } H_1 : \mu < \mu_0$$

One has to choose from the above three forms depending on the situation posed, as explained below.

In the example relating to the heights of students, the situation involved only testing the statement made about the average height of a class of students. Therefore, the Alternative Hypothesis was of the form $\mu \neq \mu_0$.

In the example about the life of electric bulbs, the company manager wanted to test whether the average life was more than 560 hrs. Therefore the Alternative Hypothesis was of the form $\mu > \mu_0$.

Consider another example, the manufacturer of cigarettes ordered that the nicotine content in a cigarette should not exceed the stipulated level of 25 mgs. In order to check this, he selected a sample of 100 cigarettes from the lot to be packed, and found that the average nicotine content was 23.5 mgs. Could he be reasonably sure that the stipulations laid down by him were being met? In this case, the manufacturer wanted the nicotine content to be less than a particular value, and therefore, the Alternative Hypothesis is of the form $\mu < \mu_0$.

Step 2 : Specification of the Level of Significance

The level of significance denoted by α is the confidence with which we reject or accept the Null hypothesis H_0 i.e., it is the maximum possible probability with which we are willing to risk an error in rejecting H_0 when it is true. The level of significance is generally specified before a test procedure so that the results obtained may not influence our decision. In practice, we take either 5% (i.e., 0.05) or 1% (i.e., 0.01) level of significance, although other levels such as 2%, 1/2% etc. may also be used. 5% Level of significance in a test procedure indicates that there are about 5 cases in 100 that we would reject the null hypothesis H_0 when it is true i.e., we are about 95% confident that we have made the right decision. Similarly, in 1% Level of significance, there is only 1 case in 100 that the null hypothesis H_0 is rejected when it is true i.e., we are about 99% confident that we have made the right decision. Level of significance is also known as the size of the test.

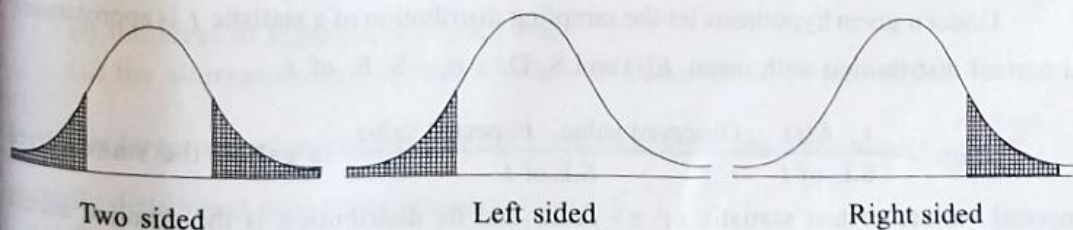
Step 3 : Identification of the Test Statistic

There are several tests of significance, viz., z, t, F etc. First we have to select the right test depending on the nature of the information given in the problem. Then we construct the test criterion and select the appropriate probability distribution.

Step 4: Critical Region: The critical region is formed based on following factors.

(a) Distribution of the Statistic i.e., whether the statistic follows the normal, 't', χ^2 or 'F' distribution (will be discussed later).

(b) **Form of Alternative Hypothesis:** If the form has $\neq$ sign, the critical region is divided equally in the left and right tails, sides of the distribution.



If the form of alternative hypothesis has $<$ sign, the entire critical region is taken in the left tail of the distribution.

If the form of alternative hypothesis has $>$ sign, the entire critical region is taken on the right side of the distribution.

Step 5 : Making Decision

We compute the value of the appropriate statistic and ascertain whether the computed value falls in acceptance or rejection region depending on the specified Level of significance. In finding the acceptance or rejection region we have to use critical values given in Statistical Tables. By comparing the computed value and the critical value decision is taken for accepting or rejecting H_0 . If the computed value $<$ critical value, we accept H_0 , otherwise we reject H_0 .

7.4 ERRORS OF SAMPLING

[JNTU (H) Dec. 2009 (Set No. 2)]

The main objective in sampling theory is to draw valid inferences about the population parameters on the basis of the sample results. In practice we decide to accept or to reject the lot after examining a sample from it. As such we have two types of errors.

(i) **Type I error** : Reject H_0 when it is true. [JNTU (A) Dec. 2009 (Set No. 3)]

If the Null Hypothesis H_0 is true but it rejected by test procedure, then the error made is called Type I error or α error.

(ii) **Type II error** : Accept H_0 when it is wrong i.e., accept H_0 when H_1 is true.

If the Null Hypothesis is false but it is accepted by test, then error committed is called Type II error or β error.

If we write

$$P(\text{Reject } H_0 \text{ when it is true}) = P(\text{Type I error}) = \alpha$$

$$\text{and } P(\text{Accept } H_0 \text{ when it is wrong}) = P(\text{Type II error}) = \beta$$

then α and β are called sizes of Type I and Type II errors respectively

$$\text{i.e., } \alpha = P(\text{Rejecting a good lot})$$

$$\beta = P(\text{Accepting a bad lot})$$

The sizes of Type I and Type II errors are also known as *producer's risk* and *consumer's risk* respectively.

The statistical testing of hypothesis aims at limiting the Type I error to a preassigned value (say : 1% or 5%) and to minimize the type II error. The only way to reduce both types of errors is to increase the sample size, if possible.

1. Critical Region

[JNTU (K) March 2014 (Set No. 1)]

Under a given hypothesis let the sampling distribution of a statistic t is approximately a normal distribution with mean $E(t)$ and S. D. $= \sigma_t = \text{S. E. of } t$.

$$\text{Then } z = \frac{t - E(t)}{\text{S.E. of } t} = \frac{\text{Observed value} - \text{Expected value}}{\text{S.E. of } t} \text{ is called the standardized}$$

normal variate or test statistic or z -score and its distribution is the standard normal distribution with mean 0 and S.D. 1 when the sample is large.

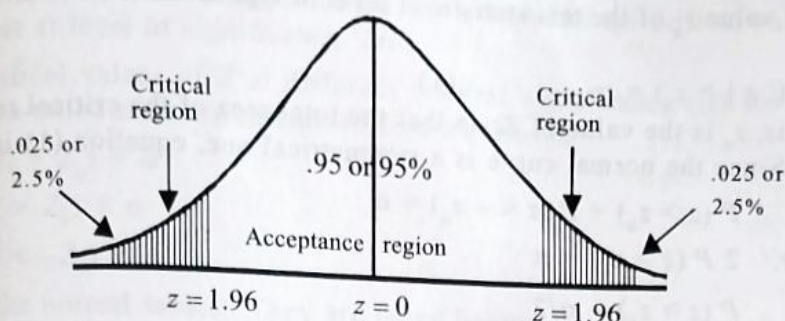


Fig.

From above figure, we see that if z lies between -1.96 and 1.96 then we are 95% confident that the hypothesis is true, since the area under the standard normal curve between $z = -1.96$ and $z = 1.96$ is 0.95 i.e., 95% of the total area i.e. $P(-1.96 \leq z \leq 1.96) = 0.95$. But if for a simple random sample we find that z lies outside the range -1.96 to 1.96 i.e., if $|z| > 1.96$, then we say that such an event occurs with probability of only 0.05 when the given hypothesis is true. In this case, we say that z -score differ significantly from the value expected under the hypothesis and hence the hypothesis H_0 is to be rejected at 5% level of significance. Thus if $|z| > 1.96$, the hypothesis H_0 is rejected at 5% level of significance. The set of z -scores outside the range -1.96 to 1.96 i.e., $|z| > 1.96$ constitutes the *critical region* or *region of rejection* of the hypothesis or the *region of significance*. On the otherhand, the set of z -scores inside the range -1.96 to 1.96 is called the region of acceptance of the hypothesis. The values -1.96 and 1.96 are called *critical values* at 5% level of significance.

Similarly, we can define critical region at any other level of significance.

Definition. A region corresponding to a statistic ' t ', in the sample space S which leads to the rejection of H_0 is called *Critical Region* or *Rejection Region*. Those region which lead to the acceptance of H_0 give us a region called *Acceptance Region*.

In other words, the region in which a sample value falling is rejected is known as the critical region. In general we take two critical regions which cover 5% and 1% areas of the normal curve.

2. Critical values or Significant values :

[JNTU 2004S]

The value of the test statistic, which separates the critical region (or rejection region) and the acceptance region is called the *critical value* or *significant value*. This value is dependent on

- (i) the level of significance used, and
- (ii) the alternative hypothesis, whether it is one-tailed or two-tailed.

For larger samples, corresponding to the statistic t , the variable $z = \frac{t - E(t)}{\text{S.E of } t}$ is normally distributed with mean 0 and variance 1.

The value of z given above under the null hypothesis is known as test statistic.

The critical value z_α of the test statistic at level of significance α for a two-tailed test is given by

$$P(|z| > z_\alpha) = \alpha \quad \dots (1)$$

That is, z_α is the value of Z so that the total area of the critical region on both tails is α . Since the normal curve is a symmetrical one, equation (1) implies,

$$P(z > z_\alpha) + P(z < -z_\alpha) = \alpha$$

$$\text{i.e. } 2 P(z > z_\alpha) = \alpha$$

$$\text{or } P(z > z_\alpha) = \alpha/2$$

That is, the area of each tail is $\alpha/2$.

The critical value z_α is that value such that the area to the right of z_α is $\alpha/2$ and the area to the left of $-z_\alpha$ is $\alpha/2$. (Refer figure).

3. Two-tailed test at level of significance ' α ' :

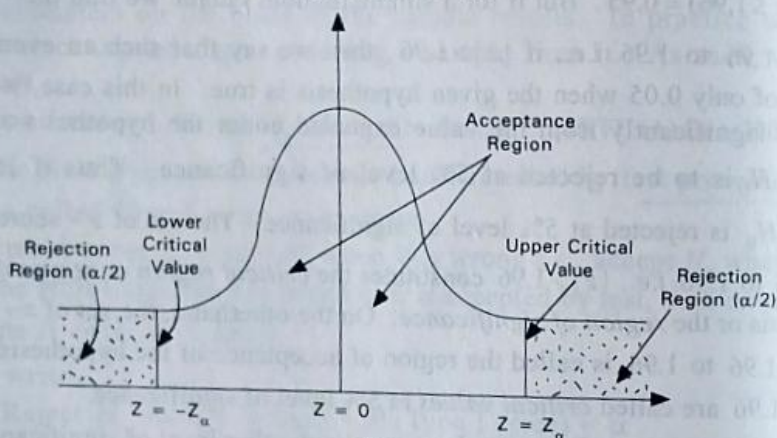


Fig.

In the case of one-tailed alternative,

$$P(Z > Z_\alpha) = \alpha \text{ if it is one-tailed (right)}$$

$$P(Z < -Z_\alpha) = \alpha \text{ if it is one-tailed (left)}$$

For Level of Significance ' α '

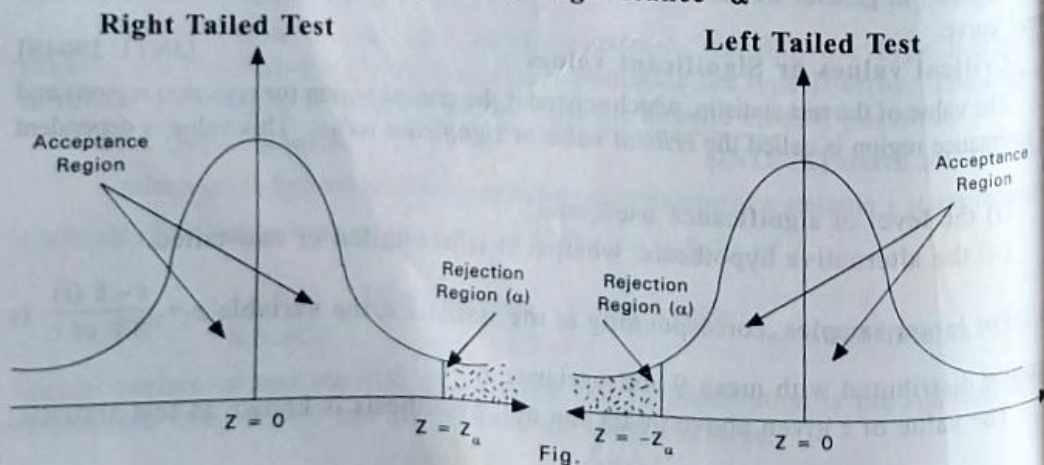


Fig.

From the above figures, it is clear, that the critical value of Z for a single-tailed test (right or left) at level of significance ' α ' is same as the critical value of Z for two-tailed test at level of significance ' 2α '.

The critical values of Z at different level of significance (α) for both single tailed and two-tailed tests are calculated from equations

$$P(|Z| > Z_\alpha) = \alpha$$

$$P(Z > Z_\alpha) = \alpha$$

$$P(Z < -Z_\alpha) = \alpha$$

using the normal tables. They are listed below.

Critical values (Z_α) of Z

	Level of Significance		
	1% (.01)	5% (.05)	10% (.1)
Two-tailed test	$ Z_\alpha = 2.58$	$ Z_\alpha = 1.96$	$ Z_\alpha = 1.645$
Right-tailed test	$Z_\alpha = 2.33$	$Z_\alpha = 1.645$	$Z_\alpha = 1.28$
Left-tailed test	$Z_\alpha = -2.33$	$Z_\alpha = -1.645$	$Z_\alpha = -1.28$

7.5 ONE - TAILED AND TWO - TAILED TESTS

[JNTU (A) Dec. 2009, (K) Nov. 2011, (H) May 2012 (Set No. 1)]

If we have to test whether the population mean μ has a specified value μ_0 , then the Null Hypothesis is $H_0 : \mu = \mu_0$ and the Alternative Hypothesis may be (i) $H_1 : \mu \neq \mu_0$ (i.e., $\mu > \mu_0$ or $\mu < \mu_0$) or (ii) $H_1 : \mu > \mu_0$ or (iii) $H_1 : \mu < \mu_0$. The Alternative Hypothesis in (i) is known as a two-tailed (i.e., both right and left tail) alternatives and the alternative hypothesis in (ii) and (iii) are known as right - tailed and left - tailed alternatives respectively.

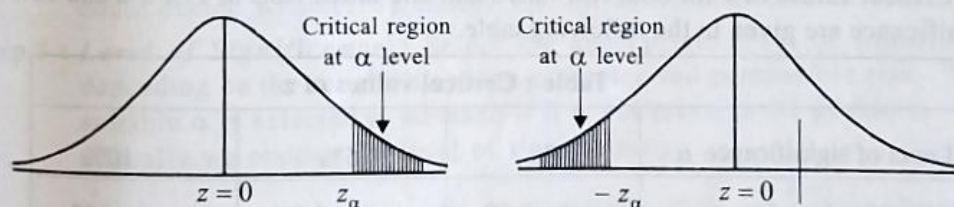


Fig.

If the Alternative Hypothesis H_1 in a test of a statistical hypothesis be one - tailed (i.e., either right - tailed or left - tailed but not both), then the test is called a *one-tailed test*. For example, to test whether the population mean $\mu = \mu_0$, we have $H_0 : \mu = \mu_0$ against the alternative hypothesis H_1 given by

(i) $H_1 : \mu > \mu_0$ (right tailed) or (ii) $H_1 : \mu < \mu_0$ (left - tailed) and the corresponding test is a single - tailed or one-tailed or one sided. In the right - tail test $H_1 : \mu > \mu_0$, the

critical region (or rejection region) $z > z_\alpha$ lies entirely in the right tail of the sampling distribution of sample mean $\bar{x}$ with area equal to the level of significance α (see figure). Similarly, in the left-tailed test ($H_1: \mu < \mu_0$), the critical region $z < -z_\alpha$ lies entirely in the left tail of the sampling distribution of the sample mean $\bar{x}$ with area equal to the level of significance α (see figure).

Two-Tailed Test

Suppose we want to test the Null Hypothesis $H_0: \mu = \mu_0$ against the Alternative Hypothesis $H_1: \mu \neq \mu_0$.

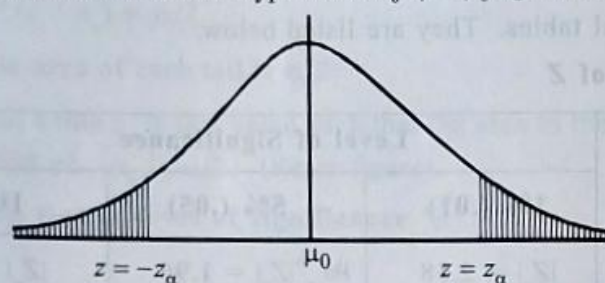


Fig.

Since H_1 is two-tailed alternative hypothesis, the critical region under the curve is equally distributed on both sides of the mean.

Thus, the critical area under the right-tail = The critical area under the left-tail
= Half of the total area

$$= \frac{1}{2} \text{ probability of rejection} = \frac{\alpha}{2}$$

with critical statistic $Z_{\alpha/2}$, where α is the level of significance.

The critical region is then, $z \leq -z_{\alpha/2}$ or $z_{\alpha/2} \leq z$.

Critical values of x for both two-tailed and one-tailed tests at 1%, 5% and 10% level of significance are given in the following table :

Table : Critical values of z			
Level of significance α	1%	5%	10%
Critical values for two-tailed test	$ z_\alpha = 2.58$	$ z_\alpha = 1.96$	$ z_\alpha = 1.645$
Critical values for Right-tailed test	$z_\alpha = 2.33$	$z_\alpha = 1.645$	$z_\alpha = 1.28$
Critical values for Left-tailed test	$z_\alpha = -2.33$	$z_\alpha = -1.645$	$z_\alpha = -1.28$

Applying one-tailed or two-tailed test for a particular problem depends entirely on the nature of the Alternative Hypothesis. If the alternative test is two-tailed we apply two-tailed test and if Alternative Hypothesis is one-tailed, we apply one-tailed test.

For example, to test whether a coin is biased or not, two - tailed test should be used, since a biased coin gives either more number of heads than tails (which corresponds to right tail) or more number of tails than heads (which corresponds to left tail)

Example : Consider two population brands of bulbs one manufactured by routine process (mean μ_1) and the other manufactured by new technique (mean μ_2). If we want to test if the bulbs differ significantly then the hypothesis is $H_0 : \mu_1 = \mu_2$ and the alternative hypothesis will be $H_0 : \mu_1 \neq \mu_2$. This gives us a two-tailed test. Suppose if we want to test if the bulbs produced by new process (μ_2) have higher average life than those produced by standard process (μ_1), then we have

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 < \mu_2$$

In this case we have to adopt a left-tailed test.

If we want to test whether the product of new process (μ_2) is inferior to that of standard process (μ_1), then we have

$$H_0 : \mu_1 = \mu_2 \text{ and } H_1 : \mu_1 > \mu_2$$

which gives a right-tailed test.

Hence the decision about applying a two-tail test or a single-tail (left or right) test will mainly depend on the problem under study.

PROCEDURE FOR TESTING OF HYPOTHESIS :

[JNTU (H) Nov. 2009 (Set No.2,4), (K) Nov. 2011, Dec. 2013 (Set No. 3)]

Various steps involved in testing of Hypothesis are given below : Infact, the same steps are followed for conducting all tests of significance.

Step 1 : Null Hypothesis : Define or set up a Null Hypothesis H_0 taking into consideration the nature of the problem and data involved.

Step 2 : Alternative Hypothesis : Set up the Alternative Hypothesis H_1 so that we could decide whether we should use one-tailed or two-tailed test.

Step 3 : Level of Significance : Select the appropriate level of significance (α) depending on the reliability of the estimates and permissible risk. That is a suitable α is selected in advance if it is not given in the problem.
(Usually we choose 5% level of significance)

Step 4 : Test Statistic: Compute the test statistic $Z = \frac{t - E(t)}{\text{S.E of } t}$ under the null hypothesis.

Here t is a sample statistic and S. E. is the standard error of t .

Step 5 : Conclusion : We compare the computed value of the test statistic Z with the critical value Z_α at given level of significance (α).

If $|Z| < Z_\alpha$, (that is, if the absolute value of the calculated value of Z is less than the critical value Z_α) we conclude that it is not significant. We accept the null hypothesis.

If $|Z| > Z_\alpha$ then the difference is significant and hence the null hypothesis is rejected at the level of significance α .

Clearly,

For two-tailed test :

If $|Z| < 1.96$ accept H_0 at 5% level of significance.

If $|Z| > 1.96$ reject H_0 at 5% level of significance.

If $|Z| < 2.58$, accept H_0 at 1% level of significance.

If $|Z| > 2.58$ reject H_0 at 1% level of significance.

For single-tailed (right or left) test :

If $|Z| < 1.645$, accept H_0 at 5% level of significance.

If $|Z| > 1.645$, reject H_0 at 5% level of significance.

If $|Z| < 2.33$ accept H_0 at 1% level of significance.

If $|Z| > 2.33$ reject H_0 at 1% level of significance.

7.6 TEST OF SIGNIFICANCE FOR LARGE SAMPLES

If the sample size $n \geq 30$, then we consider such samples as large samples. The tests of significance used in large samples are different from those used in small samples because small samples fail to satisfy the assumptions under which large sample analysis is done. If n is large, the distributions, such as Binomial, Poisson, Chi-square etc. are closely approximated by normal distributions. Therefore, for large samples, the sampling distribution of a statistic is approximately a normal distribution.

Suppose we wish to test the hypothesis that the probability of success in such trial is p . Assuming it to be true, the mean μ and the standard deviation σ of the sampling distribution of number of successes are np and $\sqrt{npq}$ respectively.

C.W.O. If x be the observed number of successes in the sample and Z is the standard normal variate then $Z = \frac{x - \mu}{\sigma}$.

Thus we have the following test of significance :

(i) If $|Z| < 1.96$, the difference between the observed and expected number of successes is not significant.

(ii) If $|Z| > 1.96$, the difference is significant at 5% level of significance.

(iii) If $|Z| > 2.58$, the difference is significant at 1% level of significance.

Assumptions for Large Samples :

The following are the assumptions under which significance tests are applied :

1. The random sampling distribution of statistic has the properties of the normal curve. This may not hold good in case of small samples.
2. Values (i.e., statistic) given by the samples are sufficiently close to the population values (i.e., parameters) and can be used in its place for calculating the standard error (S. E.) of the estimate.

For example, for computing the S. E., if the S. D. of the universe is not known, it can be replaced by the S. D. of the sample. In case of small samples, this is not possible.

Under large sample tests, we will see four important tests to test the significance.

1. Testing of significance for single proportion.
2. Testing of significance for difference of proportions.
3. Testing of significance for single mean.
4. Testing of significance for difference of means.

We will discuss Test of Significance for single proportions and difference between the proportions in the chapter "Testing of Hypothesis - II".

7.7 TEST OF SIGNIFICANCE OF A SINGLE MEAN - LARGE SAMPLES

Let a random sample of size n ($n \geq 30$) has the sample mean $\bar{x}$, and μ be the population mean. Also the population mean μ has a specified value μ_0 .

Working Rule :

1. **The Null Hypothesis** is $H_0: \bar{x} = \mu$ i.e., "there is no significance difference between the sample mean and population mean" or the sample has been drawn from the parent population.
2. **The Alternative Hypothesis** is (i) $H_1: \bar{x} \neq \mu$ ($\mu \neq \mu_0$) or (ii) $H_1: \bar{x} > \mu$ ($\mu > \mu_0$) or (iii) $H_1: \bar{x} < \mu$ ($\mu < \mu_0$)

Since n is large, the sampling distribution of $\bar{x}$ is approximately normal.

3. **Level of Significance.** Set the level of significance α .

4. **The test statistic :**

We have the following two cases.

Case I : When the standard deviation σ of population is known.

In this case, standard Error of Mean, S. E. ($\bar{x}$) = $\frac{\sigma}{\sqrt{n}}$, where n = sample size,

σ = s.d. of population.

$\therefore$ The test statistic is

$$z = \frac{\bar{x} - \mu}{\text{S.E.}(\bar{x})} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}, \text{ where } \mu \text{ is the sample mean.}$$

Case II : When the standard deviation σ of population is not known.
In this case, we take s , the s.d. of sample to compute the S. E. of mean

$$\therefore \text{S. E.}(\bar{x}) = \frac{s}{\sqrt{n}}$$

Hence the test statistic is

$$z = \frac{\bar{x} - \mu}{\text{S.E.}(\bar{x})} = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

5. Find the critical value z_α of z at the level of significance α from the normal table.

6. **Decision :**

(a) If $|z| < z_\alpha$ we accept the Null Hypothesis H_0 .

(b) If $|z| > z_\alpha$, we reject the Null Hypothesis H_0 .

The rejection rule for $H_0 : \bar{x} = \mu$ (or $\mu = \mu_0$) is given below :

Table : Critical values of z			
Level of significance α	1%	5%	10%
Critical region for $\mu \neq \mu_0$	$ z > 2.58$	$ z > 1.96$	$ z > 1.645$
Critical region for $\mu > \mu_0$	$z > 2.33$	$z > 1.645$	$z > 1.28$
Critical region for $\mu < \mu_0$	$z < -2.33$	$z < -1.645$	$z < -1.28$

Note 1 : We reject Null Hypothesis H_0 when $|z| > 3$ without mentioning any level of significance.

The test statistic is, $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ where σ is the S.D of the population

If the population S.D is not known, then use the statistic

$$z = \frac{\bar{x} - \mu}{s / \sqrt{n}} \text{ where } s \text{ is the sample S.D.}$$

2 : The values $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$ are called 95% fiducial limits or confidence limits for the mean of the population corresponding to the given sample.

Similarly, $\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$ (or) $[\bar{x} - 2.58 (\text{S.E. of } \bar{x}), \bar{x} + 2.58 (\text{S.E. of } \bar{x})]$ are called

99% confidence limits and $\bar{x} \pm 2.33 \frac{\sigma}{\sqrt{n}}$ are called 98% confidence limits.

7.8 TEST FOR EQUALITY OF TWO MEANS - LARGE SAMPLES

(TEST OF SIGNIFICANCE FOR DIFFERENCE OF MEANS OF TWO LARGE SAMPLES)

Let $\bar{x}_1$ and $\bar{x}_2$ be the sample means of two independent large random samples sizes n_1 and n_2 drawn from two populations having means μ_1 and μ_2 and standard deviations σ_1 and σ_2 . To test whether the two population means are equal.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

S. E. of $(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, where σ_1 and σ_2 are the S. D. of the two populations

To test whether there is any significant difference between $\bar{x}_1$ and $\bar{x}_2$, we have to use the statistic

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\text{S.E. of } (\bar{x}_1 - \bar{x}_2)} = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\delta = \mu_1 - \mu_2$ (=given constant.)

If $\delta = 0$, the two populations have the same means.

If $\delta \neq 0$ the two populations are different.

Under $H_0: \mu_1 = \mu_2$, the test statistic becomes
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is approximately normally distributed with mean 0 and S. D. 1.

Note : If the samples have been drawn from the population with common S. D. σ , then $\sigma_1^2 = \sigma_2^2 = \sigma^2$

$$\text{Hence } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

is normally distributed with mean zero and standard deviation one.

If σ is not known we can use an estimate of σ^2 given by $\sigma^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}$

Rejection Rule for $H_0: \mu_1 = \mu_2$

- If $|z| > 1.96$, reject H_0 at 5% level of significance.
- If $|z| > 2.58$, reject H_0 at 1% level of significance.
- If $|z| > 1.645$, reject H_0 at 10% level of significance.
- If $z > 3$ then either the samples have not been drawn from the same population or the sampling is not simple.

Otherwise accept H_0 .

Note : If the two samples are drawn from two populations with unknown Standard deviations σ_1^2 and σ_2^2 , then σ_1^2 and σ_2^2 can be replaced by sample variances S_1^2 and S_2^2 provided both the samples n_1 and n_2 are large.

In this case, the test statistic is
$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Write the formula for testing the hypothesis concerning "Two Means".

[JNTU Nov. 2008 (Set No.1)]

Solution : Let $\bar{x}_1$ be the mean of a random sample of size n_1 drawn from a population with mean μ_1 and variance σ_1^2 . Let $\bar{x}_2$ be the mean of an independent random sample of size n_2 drawn from another population with mean μ_2 and variance σ_2^2 . To test the hypothesis for difference of means, the statistic is given by

$$z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

If $|z| < 1.96$, it is accepted that the samples have come from two populations with the same mean, at 5% level of significance. Otherwise, at this level of significance, we claim that the difference in means is significant.

7.9 TEST OF SIGNIFICANCE FOR SINGLE PROPORTION - LARGE SAMPLES

[JNTU (H) Nov. 2009 (Set - 4)]

Suppose a large random sample of size n has a sample proportion p of members possessing a certain attribute (i.e., proportion of successes). To test the hypothesis that the proportion P in the population has a specified value P_0 .

Let us set the Null Hypothesis be $H_0: P = P_0$ (P_0 is a particular value of P)

Then the Alternative Hypothesis is

(i) $H_1: P \neq P_0$ (i.e., $P > P_0$ or $P < P_0$)

(ii) $H_1: P > P_0$ or

(iii) $H_1: P < P_0$

Since n is large, the sampling distribution of p is approximately normal.

$\therefore$ If H_0 is true, the test statistic

$$z = \frac{p - P_0}{\text{S.E. of } p} \text{ or } z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}, \text{ where } p \text{ is the sample proportion}$$

is approximately normally distributed.

The critical region for z depending on the nature of H_1 and level of significance α is given in the following table :

Table 7.1 : Rejection Rule for $H_0: P = P_0$			
Level of significance	1%	5%	10%
Critical region for $P \neq P_0$ (Two-tailed test)	$ z > 2.58$	$ z > 1.96$	$ z > 1.645$
Critical region for $P > P_0$ (Right-tailed test)	$z > 2.33$	$z > 1.645$	$z > 1.28$
Critical region for $P < P_0$ (Left-tailed test)	$z < -2.33$	$z < -1.645$	$z < -1.28$

Note 1 : Without any reference to the level of significance, we may reject the Null Hypothesis H_0 when $|z| > 3$.

Note 2 : (i) Limits for population proportion P are given by $p \pm 3\sqrt{\frac{pq}{n}}$ where

$$q = 1 - p.$$

(ii) Confidence interval for proportion P for large sample at α level of significance is

$$P - z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} < P < P + z_{\alpha/2} \cdot \sqrt{\frac{PQ}{n}} \text{ where } Q = 1 - P \text{ and}$$

$$z_{\alpha/2} = 1.96 \text{ (for 95\%), } z_{\alpha/2} = 2.33 \text{ (for 98\%) and}$$

$$z_{\alpha/2} = 2.58 \text{ (for 99\%).}$$

7.10 TEST FOR EQUALITY OF TWO PROPORTIONS (OR TEST OF SIGNIFICANCE OF DIFFERENCE BETWEEN TWO SAMPLE PROPORTIONS - LARGE SAMPLES)

Let p_1 and p_2 be the sample proportions in two large random samples of sizes n_1 and n_2 drawn from two populations having proportions P_1 and P_2 .

To test whether the two samples have been drawn from the same population.

1. The Null Hypothesis $H_0 : P_1 = P_2$

2. The Alternative Hypothesis $H_1 : P_1 \neq P_2$

3. The Test Statistic : There are two ways of computing a test statistic z .

(a) When the population proportions P_1 and P_2 are known.

In this case, $Q_1 = 1 - P_1$ and $Q_2 = 1 - P_2$ and p_1, p_2 are sample proportions.

$$\therefore \text{Standard Error of Difference} = \text{S. E. } (p_1 - p_2) = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$

(b) When the population proportions P_1 and P_2 are not known but sample proportions p_1 and p_2 are known.

In this case we have two methods to estimate P_1 and P_2 .

(i) Method of Substitution : In this method, sample proportions p_1 and p_2 are substituted for P_1 and P_2 .

$$\therefore \text{S.E.}(p_1 - p_2) = \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$$

(ii) Method of Pooling : In this method, the estimated value for the two population proportions is obtained by pooling the two sample proportions p_1 and p_2 into a single proportion p by the formula given below.

Sample proportion of two samples or estimated value of p is given by

$$p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, \text{ so that } q = 1 - p$$

$$\therefore \text{S.E.}(p_1 - p_2) = \sqrt{\frac{pq}{n_1} + \frac{pq}{n_2}} = \sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Hence the test statistic is

$$z = \frac{p_1 - p_2}{\text{S.E.}(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Rejection Rule for $H_0 : P_1 = P_2$.

- (i) If $|z| > 1.96$, then reject H_0 at 5% level of significance.
- (ii) If $|z| > 2.58$, then reject H_0 at 1% level of significance.
- (iii) If $|z| > 1.645$, then reject H_0 at 10% level of significance.

TEST OF SIGNIFICANCE

(SMALL SAMPLES)

8.1 INTRODUCTION

In the earlier chapter, we considered certain tests of significance based on the theory of the normal distribution. The assumptions made in deriving those tests will be valid only for large samples. When the sample is small ($n < 30$), we can use normal distribution to test for a specified population mean or difference of two population means as in large sample tests only when the sample is drawn from a normal population whose S.D., σ is known. If σ is not known, we cannot proceed as above. If a population is normally distributed, the sampling distribution of the sample mean for any sample size is also normally distributed whether σ is known or not.

8.2 DEGREE OF FREEDOM (d.f.)

[JNTU (K) Dec. 2013 (Set No. 2)]

The number of independent variates which make up the statistic is known as the degree of freedom (d.f.) and it is denoted by ν (the letter 'Nu' of the Greek alphabet). In other words, it is the number of values in a set of data which may be assigned arbitrarily or, it refers to the number of "independent constraints" in a set of data.

Definition : It is a number which indicates how many of the values of a variable may be independently (or freely) chosen. For example, if we have to choose any four numbers freely, then we may choose 11, 6, 14, 28 or any other set of four numbers. As in this case all the four numbers have freedom to vary, we say that the degrees of freedom is 4. If we impose a restriction on the numbers, say the sum is 50, then we can choose first three numbers freely and the fourth number is such that the sum is 50. Thus, the three variables are free and independent choices for finding the fourth. Hence these are the degrees of freedom. In this case, the degrees of freedom is 3, i.e., $4 - 1$.

In general, the number of degrees of freedom is equal to the total number of observations less the number of independent constraints imposed on the observations. For example, in a set of data of n observations, if k is the number of independent constraints then $\nu = n - k$.

8.3 t-DISTRIBUTION (OR) STUDENT'S t-DISTRIBUTION

[JNTU (H) Dec. 2011 (Set No. 1)]

It is used for testing of hypothesis when the sample size is small and population S.D. σ is not known.

Definition : If $\{x_1, x_2, \dots, x_n\}$ be any random sample of size n drawn from a normal (or approximately normal) population with mean μ and variance σ^2 , then the

test statistic t is defined by $t = \frac{\bar{x} - \mu}{S / \sqrt{n}}$ where $\bar{x}$ = sample mean and

$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is an unbiased estimate of σ^2 . The test statistic $t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$ is a random variable having the t -distribution with $\nu = n - 1$ degrees of freedom and with probability density function $f(t) = y_0 \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$ where $\nu = n - 1$ and y_0 is a constant got by $\int_{-\infty}^{\infty} f(t) dt = 1$. This is known as "Student's t -distribution" or simply " t distribution".

The above result is more general than the central limit theorem, since it does not require the knowledge of σ , and since population is assumed to be normal it is less general than the central limit theorem.

Properties of t -Distribution :

[JNTU (A) Nov. 2011, (H) Dec. 2011, (K) Dec. 2013 (Set No.2)]

1. The shape of t -distribution is **bell-shaped**, which is similar to that of a normal distribution and is symmetrical about the mean.

2. The t -distribution curve is also asymptotic to the t -axis, i.e., the two tails of the curve on both sides of $t=0$ extends to infinity.

3. It is symmetrical about the line $t=0$.

4. The form of the probability curve varies with degrees of freedom i.e., with sample size.

5. It is unimodal with Mean = Median = Mode.

6. The *mean* of standard normal distribution and as well as t -distribution is zero but the variance of t -distribution depends upon the parameter ν which is called the degrees of freedom.

7. The variance of t -distribution exceeds 1, but approaches 1 as $n \rightarrow \infty$. Infact the t -distribution with ν -degrees of freedom approaches standard normal distribution as $\nu = (n-1) \rightarrow \infty$.

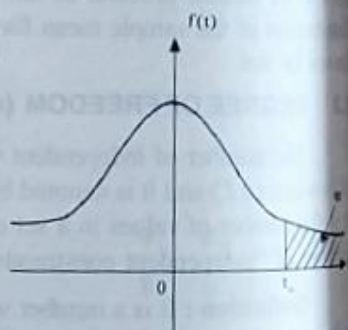
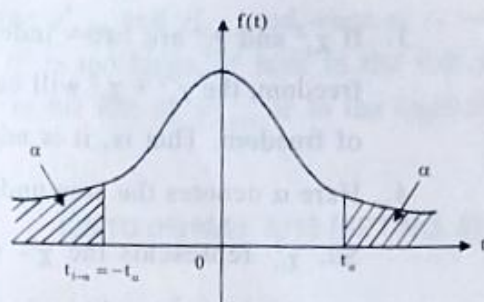


Fig. t -distribution

The selected values of t_α for various values of ν can be obtained from the tables of t -distribution, where t_α denotes the area under t -distribution to its right is equal to α . These values are tabulated in tables. In tables the left-hand column contains values of ν , the column heading are areas α in the right-hand tail of the t -distribution, and the entries are values of t_α . It is not necessary to tabulate values of t_α for $\alpha < 0.50$, as it follows from the symmetry of the t -distribution that $t_{1-\alpha} = -t_\alpha$ i.e., the t -value leaving an area of $1-\alpha$ to the right and therefore an area α to its left, is equal to the negative t -value which leads an area α in the right. The bottom row of the table of t -distribution correspond to the values of Z that cut off right hand tails of area α under the standard normal curve. We use the notation Z_α for such a value of Z , it can be seen for example, that $Z_{0.025} = 1.96 = t_{0.025}$ for $\nu = \infty$.

This result should really have been expected, since the t -distribution approaches the standard normal distribution as $v \rightarrow \infty$. In fact, if we observe, we find that the value of t_α for 29 or more degrees of freedom are close to the corresponding values of Z ; we conclude that the standard normal distribution provides a good approximation to the t -distribution for samples of size $n \geq 30$.

Fig. Symmetry of t -distribution

The t -distribution is extensively used in hypothesis about one mean, or about equality of two means when σ is unknown.

8.4. APPLICATIONS OF THE t - DISTRIBUTION

The t - distribution has a wide number of applications in Statistics, some of them are given below :

- (1) To test the significance of the sample mean, when population variance is not given
- (2) To test the significance of the mean of the sample *i.e.*, to test if the sample mean differs significantly from the population mean.
- (3) To test the significance of the difference between two sample means or to compare two samples.
- (4) To test the significance of an observed sample correlation coefficient and sample regression coefficient.

8.5 CHI-SQUARE (χ^2) DISTRIBUTION

[JNTU (H) May 2012 (Set No. 3), (K) Dec. 2013 (Set No. 3)]

Chi-squared distribution is a continuous probability distribution of a continuous random variable X with probability density function given by

$$f(x) = \begin{cases} \frac{1}{2^{v/2} \Gamma(v/2)} x^{(v/2)-1} \cdot e^{-x/2}, & \text{for } x > 0 \\ 0, & \text{otherwise} \end{cases}$$

where v is a positive integer is the only single parameter of the distribution, also known as "degrees of freedom (dof)".

χ^2 - distribution was extensively used as a measure of goodness of fit and to test the independence of attributes.

Properties of χ^2 - distribution :

1. χ^2 - distribution curve is not symmetrical, lies entirely in the first quadrant, and hence not a normal curve, since χ^2 varies from 0 to ∞ .
2. It depends only on the degrees of freedom v .

3. If χ_1^2 and χ_2^2 are two - independent distributions with v_1 and v_2 degrees of freedom, the $\chi_1^2 + \chi_2^2$ will be chi-squared distribution with $(v_1 + v_2)$ degrees of freedom. That is, it is additive.

4. Here α denotes the area under the chi-square distribution to the right of χ_α^2 .

So, χ_α^2 represents the χ^2 - value such that the area under the chi-square curve to its (χ_α^2) right is equal to α . Chi-square distribution is very important in estimation and hypothesis testing. χ^2 -distribution is used in sampling distribution, analysis of variance, mainly, it is used as a measure of goodness of fit and in analysis of $r \times c$ tables.

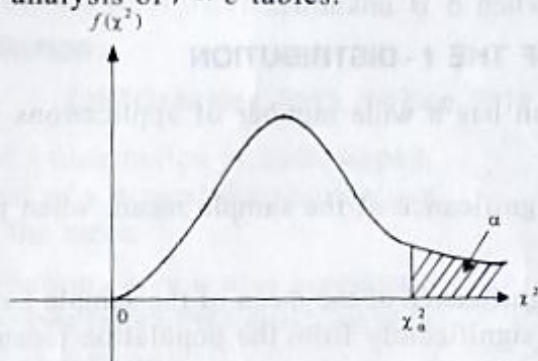


Fig. χ^2 - distribution

For various values of α and v , the values of χ_α^2 are tabulated in tables.

In χ^2 table, the left hand column contains values of v (degrees of freedom), the column headings are areas α in the right hand tail of χ^2 - distribution curve, the entries are χ^2 values. It is necessary to calculate values of χ_α^2 for $\alpha > 0.50$ since χ^2 curve or distribution is not symmetrical.

5. Mean = v and variance = $2v$

Sampling distribution of variance S^2 :

The theoretical sampling distribution of the sample variance for random samples from normal population is related to the chi-squared distribution as follows :

Let S^2 be the variance of a random sample of size n , taken from a normal population having the variance σ^2 .

$$\text{Then } \chi^2 = \frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\sigma^2} \left[\because S^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1} \right]$$

is a value of a random variable having the χ^2 -distribution with $v = n - 1$ degrees of freedom.

Exactly 95% of χ^2 -distribution lies between $\chi^2_{0.975}$ and $\chi^2_{0.025}$ and when σ^2 is too small, χ^2 falls to the right of $\chi^2_{0.025}$ and when σ^2 is too large, χ^2 falls to the left of $\chi^2_{0.975}$. Thus when σ^2 is correct, χ^2 -values are to the left of $\chi^2_{0.975}$ or to the right of $\chi^2_{0.025}$.

8.6 APPLICATIONS OF χ^2 DISTRIBUTION [JNTU (H) MAY 2012 (SET NO. 3)]

- (1) To test the goodness of fit.
- (2) To test the independence of attributes.
- (3) To test the homogeneity of independent estimation of the population variance.
- (4) To test the homogeneity of independent estimation of the population correlation coefficient.

8.7 F-DISTRIBUTION

(SAMPLING DISTRIBUTION OF THE RATIO OF TWO SAMPLE VARIANCES)

Another important continuous probability distribution which plays an important role in connection with sampling from normal populations is the F-distribution.

Let S_1^2 be the sample variance of an independent sample of size n_1 drawn from a normal population $N(\mu_1, \sigma_1^2)$. Similarly, let S_2^2 be the sample variance in an independent sample of size n_2 drawn from another normal population $N(\mu_2, \sigma_2^2)$.

Thus S_1^2 and S_2^2 are two variances of two random samples of sizes n_1 and n_2 respectively drawn from the normal population, with the variances σ_1^2 and σ_2^2 . To determine whether the two samples come from two populations having equal variances.

Consider the sampling distribution of the ratio of the variances of the two independent random samples defined by

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}$$

which follows F-distribution with $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ degrees of freedom.

F-distribution can be used to test the equality of several population means, comparing sample variances and analysis of variance.

Under the assumption (hypothesis) that two normal populations have the same variance i.e., $\sigma_1^2 = \sigma_2^2$, we have $F = \frac{S_1^2}{S_2^2}$

F determines whether the ratio of two sample variances S_1 and S_2 is too small or too large.

When F is close to 1, the two sample variances S_1 and S_2 are almost same. In practice, it is customary, to take the larger sample variance as the numerator.

F is always a positive number.

The sampling distribution of F is of the form $f(F) = K \frac{F^{(v_1-2)/2}}{(v_1 F + v_2)^{(v_1+v_2)/2}}$ where

v_1 and v_2 are two degrees of freedom and K is determined by $\int_0^{\infty} f(F) dF = 1$.

Properties of F -distribution :

- (i) F - distribution is free from population parameters and depends upon degrees of freedom only.
- (ii) F -distribution curve lies entirely in first-quadrant.
- (iii) The F -curve depends not only on the two parameters v_1 and v_2 but also on the order in which they are stated.

$$(iv) F_{1-\alpha}(v_1, v_2) = \frac{1}{F_{\alpha}(v_2, v_1)}$$

where $F_{\alpha}(v_1, v_2)$ is the value of F with v_1 and v_2 degrees of freedom such that the area under the F -distribution curve to the right of F_{α} is α .

- (v) The mode of F - distribution is less than unity.

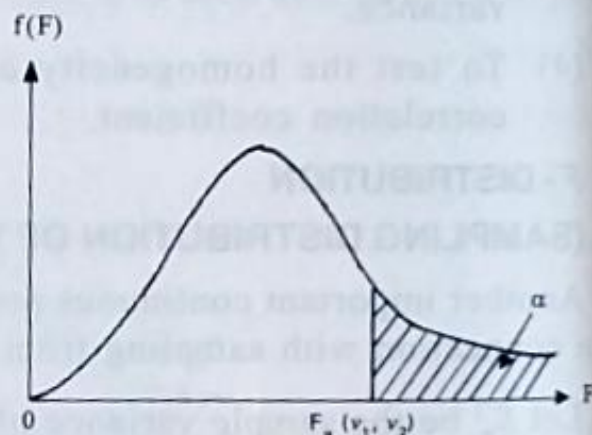


Fig. F - distribution

8.8 TEST OF SIGNIFICANCE FOR SMALL SAMPLES

A very important aspect of the sampling theory is the study of tests of significance, which enable us to decide on the basis of the sample results, if

- (i) The deviation between the observed sample statistic and the hypothetical parameter value is significant.
- (ii) The deviation between two sample statistics is significant.

The following are some important tests for small samples :

- (i) Student's 't' test
- (ii) F-test
- (iii) χ^2 -test

8.9 STUDENT'S 't' TEST

Let $\bar{x}$ = Mean of a sample

n = Size of the sample

σ = Standard deviation of the sample

μ = Mean of the population supposed to be normal.

Then the student's t is defined by the statistic $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$

If s^2 be the sample variance, $s^2 = \frac{\sum (x_i - \bar{x})^2}{n}$

If $S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$, then $\frac{S^2}{s^2} = \frac{n}{n-1}$ or $S = \sqrt{\frac{n}{n-1}} \cdot s$

$$\therefore t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

S^2 is called the unbiased estimate of population variance σ^2 .

Note : 1. If the standard deviation of a sample is given directly, then the statistic is

$$\text{given by } t = \frac{\bar{x} - \mu}{\text{S.D.}/\sqrt{n-1}}$$

2. If the calculated value of t exceeds the table value of t at 5% level of significance, the null hypothesis H_0 is rejected. If the calculated value of t is less than the tabulated value of t at 5% level the null hypothesis is accepted.

Assumptions for Student's t -test :

The following are made in student's t test

- (1) Sample size, $n < 30$
- (2) The parent population from which sample is drawn is normal.
- (3) The population standard deviation is unknown.
- (4) The sample observations are independent, i.e., sample is random.

Uses of t -test :

This test is used

- (1) to test for a specified mean
- (2) to test for equality of two means of two independent samples drawn from two normal populations, S.D. of the populations being unknown, and
- (3) to test the significance of difference between the means of paired data.

8.10 CONFIDENCE OR FIDUCIAL LIMITS FOR μ

Suppose we want to find from the sample data the limits within which the population mean will lie with a probability of 0.95. The limits are called the 95% confidence limits of the population mean for the given sample.

If $t_{0.05}$ is the table value of t for $(n-1)$ degrees of freedom at 5% level of significance, then 95% confidence limits for μ are given by $\bar{x} \pm t_{0.05} \cdot \frac{S}{\sqrt{n}}$.

For $P(|t| > t_{0.05}) = 0.05$ i.e., $P(|t| \leq t_{0.05}) = 0.95$

95% confidence limits for μ are given by

$$|t| \leq t_{0.05} \quad \text{i.e.,} \quad \left| \frac{\bar{x} - \mu}{S/\sqrt{n}} \right| \leq t_{0.05} \quad \text{or} \quad \frac{|\bar{x} - \mu|}{S/\sqrt{n}} \leq t_{0.05}$$

$$\Rightarrow -t_{0.05} \leq \frac{\bar{x} - \mu}{S/\sqrt{n}} \leq t_{0.05}$$

$$\Rightarrow \bar{x} - t_{0.05} \cdot S/\sqrt{n} \leq \mu \leq \bar{x} + t_{0.05} \cdot S/\sqrt{n}$$

Similarly 99% confidence limits for μ are $\bar{x} \pm t_{0.01} \cdot S/\sqrt{n}$

where $t_{0.01}$ is the tabulated value of t for $(n-1)$ degrees of freedom at 1% level of significance.

8.11 STUDENT'S 't' TEST FOR SINGLE MEAN

Suppose we want to test

- if a random sample x_i of size n has been drawn from a normal population with a specified mean μ .
- if the sample mean differs significantly from the hypothetical value μ of the population mean.

In this case the statistic is given by $t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$ where $\bar{x}$, μ , S , n have usual meanings.

Let a random sample of size n ($n < 30$) has a sample mean $\bar{x}$. To test the hypothesis that the population mean μ has a specified value μ_0 when population S. D. σ is not known.

Let the Null Hypothesis be $H_0: \mu = \mu_0$

Then the Alternative Hypothesis is $H_1: \mu \neq \mu_0$.

Assuming that H_0 is true, the test statistic given by $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$, where s is the sample S. D. follows t -distribution with $v = (n-1)$ d.f.

We calculate the value of $|t|$ and compare this value with the table value of t at α level of significance. If the calculated value of $t >$ the table value of t , we reject H_0 at α level. Otherwise we accept H_0 .

In this case, 95% confidence limits for the population mean μ are $\bar{x} \pm t_{\alpha} \cdot \frac{s}{\sqrt{n-1}}$

where $\alpha = 0.025$ for two-tailed test and $s =$ sample S. D. and 99% confidence limits for μ are $\bar{x} \pm t_{\alpha} \cdot \frac{s}{\sqrt{n-1}}$ where $\alpha = 0.05$.

For a two-tailed test at α level of significance, value of $\alpha/2$ is taken for α .

8.12 STUDENT'S t - TEST FOR DIFFERENCE OF MEANS

Let $\bar{x}$ and $\bar{y}$ be the means of two independent samples of sizes n_1 and n_2 ($n_1 < 30, n_2 < 30$) drawn from two normal populations having means μ_1 and μ_2 . To test whether the two population means are equal (i.e., to test whether the difference $\mu_1 - \mu_2$ is significant).

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

If $\sigma_1 = \sigma_2 = \sigma$, then an unbiased estimate S^2 of the common variance σ^2 is given by $S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$ where s_1^2, s_2^2 are the two sample variances.

Also S. E. of $(\bar{x} - \bar{y}) = S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ where $S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$

Assuming that H_0 is true, the test statistic t given by $t = \frac{\bar{x} - \bar{y}}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$, follows t -distribution with $(n_1 + n_2 - 2)$ d.f.

Here $\bar{x} = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i$, $\bar{y} = \frac{1}{n_2} \sum_{i=1}^{n_2} y_i$ and $S^2 = \frac{1}{n_1 + n_2 - 2} [\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2]$

or $S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$

We calculate the value of $|t|$ and compare this value with the table value at α level of significance. If the calculated value of $|t| >$ the table value, we reject H_0 at α level. Otherwise we accept H_0 .

95% confidence limits for the difference of two population means are

$$(\bar{x} - \bar{y}) \pm t_{\alpha} \cdot S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad \text{where } \alpha = 0.025.$$

and 99% confidence limits for the difference of two population means are

$$(\bar{x} - \bar{y}) \pm t_{\alpha} \cdot S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \quad \text{where } \alpha = 0.0005$$

8.13 PAIRED - SAMPLE t - TEST

Paired observations arise in many practical situations where each homogeneous experimental unit receives both population conditions. As a result, each experimental unit has a pair of observations, one for each population.

For instance, to test the effectiveness of "drug" some 11 persons blood pressure is measured "before" and "after" the intake of certain drug. Here the individual person is the experimental unit and the two populations are blood pressure "before" and "after" the drug is given. Thus for each observation in one sample, there is a corresponding observation in the other sample pertaining to the same character. Hence the two samples are not independent.

Consider another example. Suppose a business concern is interested to know whether a particular media of promoting sales of a product is really effective or not. In this case we have to test whether the average sales before and after the sales promotion are equal.

If $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ be the pairs of sales data before and after the sales promotion in a business concern, we apply paired t - test to examine the significance of the difference of the two situations.

Let $d_i = x_i - y_i$ or $y_i - x_i$ for $i = 1, 2, 3, \dots, n$

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$ (i.e. $\mu = 0$), there is no significant difference between the means in two situations.

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Assuming that H_0 is true, the test statistic for n paired observations (which are dependent) by taking the differences $d_1, d_2, \dots, d_n$ of the paired data.

$$t = \frac{\bar{d} - \mu}{S / \sqrt{n}} = \frac{\bar{d}}{S / \sqrt{n}} \quad (\because \mu = 0)$$

where $\bar{d} = \frac{1}{n} \sum d_i$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2$

$$\text{or } S^2 = \frac{\sum d^2 - n(\bar{d})^2}{n-1} \quad \text{or } \frac{1}{n-1} \left[\sum d^2 - \frac{(\sum d)^2}{n} \right]$$

are the mean and variance of the differences $d_1, d_2, \dots, d_n$ respectively and μ is the mean of the population of differences.

The above statistic follows student's t - distribution with $(n-1)$ degrees of freedom.

When testing the significance of the difference of the means of two samples, we assumed that the two samples came from the same population or from populations with equal variances. If the variances of the populations are not equal, a significant difference in the means may arise. Hence, before we apply the t -test for the significance of the difference of two means, we have to test for the equality of population variances using F -test of significance.

If s_1^2 and s_2^2 are the variances of two samples of sizes n_1 and n_2 respectively, then the population variances are given by

$$n_1 s_1^2 = (n_1 - 1) S_1^2 \text{ and } n_2 s_2^2 = (n_2 - 1) S_2^2$$

The quantities $v_1 = n_1 - 1$ and $v_2 = n_2 - 1$ are called degrees of freedom of these estimates. We want to test if these estimates S_1^2 and S_2^2 are significantly different or if the samples may be regarded as drawn from the same population or from two populations with same variance σ^2 .

Test for Equality of Two population variances

Let two independent random samples of sizes n_1 and n_2 be drawn from two normal populations.

To test the hypothesis that the two population variances σ_1^2 and σ_2^2 are equal.

Let the Null Hypothesis be $H_0 : \sigma_1^2 = \sigma_2^2$

Then the Alternative Hypothesis is $H_1 : \sigma_1^2 \neq \sigma_2^2$

The estimates of σ_1^2 and σ_2^2 are given by

$$S_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{\sum (x_i - \bar{x})^2}{n_1 - 1} \text{ and } S_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{\sum (y_i - \bar{y})^2}{n_2 - 1}, \text{ where } s_1^2 \text{ and } s_2^2 \text{ are the}$$

variances of the two samples.

Assuming that H_0 is true, the test statistic $F = \frac{S_1^2}{S_2^2}$ or $\frac{S_2^2}{S_1^2}$ according as $S_1^2 > S_2^2$ or

$S_2^2 > S_1^2$ follows F -distribution with $(n_1 - 1, n_2 - 1)$ degrees of freedom.

Conclusion : If the calculated value of $F >$ the tabulated value of F at 5% level, we reject the Null Hypothesis H_0 and conclude that the variances σ_1^2 and σ_2^2 are not equal. Otherwise, we accept the Null Hypothesis H_0 and conclude that σ_1^2 and σ_2^2 are equal.

Note : 1. In numerical problems, we take the greater of the two variances S_1^2, S_2^2 in the numerator and the other in the denominator.

$$\text{i.e., } F = \frac{\text{Greater variance}}{\text{Smaller variance}}$$

2. When F is close to 1, the two sample variances S_1 and S_2 are nearly same.
3. If sample variance S^2 is given, we can obtain population variance σ^2 by using the relation $n \sigma^2 = (n - 1) S^2$ and vice-versa.
4. $F_\alpha(v_1, v_2)$ is the value of F with v_1 and v_2 degrees of freedom such that the area under the F - distribution to the right of F_α is α . In tables F_α tabulated for $\alpha = 0.05$ and $\alpha = 0.01$ for various combinations of the d.o.f v_1 and v_2 . Clearly value of F at 5% significance is lower than at 1%

8.15 CHI-SQUARE (χ^2) TEST

When a fair coin is tossed 100 times, the theoretical considerations leads us to expect 50 heads and 50 tails. But in practice, these results are rarely achieved i.e., the results obtained in an experiment do not agree exactly with the theoretical results. The magnitude of discrepancy between the theory and observation is given by the quantity χ^2 (a Greek letter, pronounced as "chi-square"). If $\chi = 0$, the observed and expected frequencies completely coincide. As the value of χ^2 increases, the discrepancy between the observed and theoretical frequencies increases. Thus, χ^2 affords a measure of the correspondence between theory and observation.

Definition : If a set of events $A_1, A_2, \dots, A_n$ are observed to occur with frequencies $O_1, O_2, \dots, O_n$ respectively and according to probability rules $A_1, A_2, \dots, A_n$ are expected to occur with frequencies $E_1, E_2, \dots, E_n$ respectively with $O_1, O_2, \dots, O_n$ are called observed frequencies and $E_1, E_2, \dots, E_n$ are called expected frequencies.

If O_i ($i=1, 2, \dots, n$) is a set of observed (experimental) frequencies and E_i ($i=1, 2, \dots, n$) is the corresponding set of expected (theoretical) frequencies, then

χ^2 is defined as $\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$ with $(n-1)$ degrees of freedom.

χ^2 is used to test whether differences between observed and expected frequencies are significant.

Note : If the data is given in a series of ' n ' numbers then degrees of freedom = $n - 1$.

In case of Binomial distribution, d.f. = $n - 1$.

In case of Poisson distribution, d.f. = $n - 2$

In case of Normal distribution, d.f. = $n - 3$

Chi-square Distribution is an important continuous probability distribution and it is used in both large and small tests. In chi-square tests, χ^2 - distribution is mainly used

(i) to test the goodness of fit,

(ii) to test the independence of attributes,

(iii) to test if the population has a specified value of the variance σ^2 .

We shall discuss these tests one by one as applications of χ^2 - distribution.

8.16 χ^2 TEST AS A TEST OF GOODNESS OF FIT

We use this test to decide whether the discrepancy between theory and experiment is significant or not i.e., to test whether the difference between the theoretical and observed values can be attributed to chance or not.

Let the Null Hypothesis H_0 be that there is no significant difference between the observed values and the corresponding expected values.

The Alternative Hypothesis H_1 is that the above difference is significant.

Let $O_1, O_2, \dots, O_n$ be a set of observed frequencies and $E_1, E_2, \dots, E_n$ the corresponding set of expected frequencies. Then the test statistic χ^2 is given by

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right] = \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \dots + \frac{(O_n - E_n)^2}{E_n}$$

Assuming that H_0 is true, the test statistic χ^2 follows Chi-square distribution with $(n-1)$ d.f., where

$$\sum_{i=1}^n O_i = \sum_{i=1}^n E_i \text{ or } \sum_{i=1}^n (O_i - E_i) = 0$$

Conclusion : If the calculated value of $\chi^2 >$ tabulated value of χ^2 at α level, the Null Hypothesis H_0 is rejected. Otherwise, H_0 is accepted.

χ^2 test enables us to ascertain how well the theoretical distributions such as Binomial, Poisson, Normal, etc., fit the distributions obtained from sample data.

8.17 CONDITIONS OF VALIDITY

Following are the conditions which should be satisfied before χ^2 test can be applied.

1. The sample observations should be independent.
2. N , the total frequency is large, i.e., > 50 .
3. The constraints on the cell frequencies, if any, are linear.
4. No theoretical (or expected) frequency should be less than 10. If small theoretical frequencies occur, the difficulty is overcome by grouping 2 or more classes together before calculating $(O-E)$. Note that the degrees of freedom is determined with the number of classes after regrouping.

8.18 CHI-SQUARE TEST FOR INDEPENDENCE OF ATTRIBUTES

Definition : Literally, an attribute means a quality or characteristic. Examples of attributes are drinking, smoking, blindness, honesty, beauty etc.

An attribute may be marked by its presence (position) or absence in a number of a given population. Let the observations be classified according to two attributes and the frequencies O_i in the different categories be shown in a two-way table, called contingency table. We have to test on the basis of cell frequencies whether the two attributes are independent or not. We take the Null - Hypothesis H_0 that there is no association between the attributes *i.e.*, we assume that the two attributes are

independent. The expected frequencies (E_i) of any cell = $\frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$

The test statistic $\chi^2 = \sum_i \left[\frac{(O_i - E_i)^2}{E_i} \right]$ approximately follows Chi-square distribution with d.f. = (No. of rows - 1) $\times$ (No. of columns - 1)

If the calculated value of χ^2 is less than the table value at a specified level (generally 5%) of significance, the hypothesis holds good *i.e.*, the attributes are independent and do not bear any association. On the other hand, if the calculated value of χ^2 is greater than the table value at a specified level of significance, we say that the results of the experiment do not support the hypothesis, in other words, the attributes are associated.

Let us consider two attributes A and B . A is divided into two classes and B is divided into two classes. The various cell frequencies can be expressed in the following table known as 2×2 contingency table.

A	a	b
B	c	d
a	b	$a + b$
c	d	$c + d$
$a + c$	$b + d$	$N = a + b + c + d$

The expected frequencies are given by

$E(a) = \frac{(a+c)(a+b)}{N}$	$E(b) = \frac{(b+d)(a+b)}{N}$	$a + b$
$E(c) = \frac{(a+c)(c+d)}{N}$	$E(d) = \frac{(b+d)(c+d)}{N}$	$c + d$
$a + c$	$b + d$	$N = a + b + c + d$

The value of χ^2 is given by $\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$ where

$N = a + b + c + d$ with d.f. $= (2-1)(2-1) = 1$. We use this formula when the expected frequencies are in fractions (or decimals).

8.19 CHI-SQUARE TEST FOR POPULATION VARIANCE

Suppose that a random sample x_i ($i = 1, 2, \dots, n$) is drawn from a normal population with mean μ and variance σ^2 . To test the hypothesis that the population variance σ^2 has a specified value σ_0^2 .

Let the Null Hypothesis be $H_0: \sigma^2 = \sigma_0^2$ (or $\sigma = \sigma_0$)

Then the Alternative Hypothesis is $H_1: \sigma^2 \neq \sigma_0^2$ (or $\sigma^2 > \sigma_0^2$)

The test statistic is $\chi^2 = \sum \frac{(x_i - \bar{x})^2}{\sigma_0^2} = \frac{ns^2}{\sigma_0^2}$

where $s^2 = \text{sample variance} = \frac{\sum (x_i - \bar{x})^2}{n}$ and $ns^2 = (n-1)S^2$

Assuming that H_0 is true, the test statistic χ^2 follows Chi-square distribution with $(n-1)$ d.f.

Conclusion : If the calculated value of $\chi^2 >$ the tabulated value, we reject the Null Hypothesis H_0 . Otherwise, we accept H_0 .

8.20 ANALYSIS OF VARIANCE (ANOVA) - A Test for Homogeneity of Mean

We know that variance of a series of observations is the mean (A.M.) of the squares of the differences of the observations from their mean, and variance is used as a measure of variation. Student's t - test is used to test whether there is any significant difference (or variation) in the means of two populations. This test fails when we consider the means of three or more populations. The equality of several population means can be tested by comparing the sample variances using F-distribution *i.e.*, by analysis of variance. This is one of the most

elegant and versatile statistical techniques and finds wide applications in determining whether the difference between the means of three or more populations is significant or not. The systematic procedure of this statistical technique was first developed by Prof. R. A. Fisher, an Agriculturist who made an extensive use of this technique in his agriculture experiments. Now it is widely used in professional business and physical sciences. For example, the differences in various types of drugs manufactured by a company to cure a particular disease may be studied through this statistical technique.

In this chapter our main aim is to test on the basis of sample observations whether the means of three or more populations are equal or not. The technique of analysis of variance is referred to as ANOVA. The technique of "ANOVA" is to split the variation into its various components. Usually the variance (or total variance) is splitted into the following two parts :

- (i) Variance between samples
- (ii) Variance within samples.

Assumptions in Analysis of Variance

The analysis of variance is based on the following assumptions :

1. The samples are independently (or randomly) drawn from the populations.
2. Populations from which the samples are selected are normally distributed.
3. Each of the populations has the same variance.

If, however, the sample sizes are large enough, we do not need the assumption of normality.

A table showing the source of variation, the sum of squares, degrees of freedom, mean squares (variance), and the formula for the F-ratio is known as ANOVA table.

8.21 THE TECHNIQUE OF ANALYSIS OF VARIANCE

The observations (or data) may be classified according to one factor (or criterion) or two factors. The classifications according to one-factor and two-factors are called **one-way classification** and **two-way classification** respectively.

ANOVA for one-way classification

In the investigation of analysis of variance (ANOVA), if we consider the influence of any one factor, then it is called one-way classification. For example, the yields of several plots of land may be classified according to one or more types of fertilizers.

The techniques for ANOVA one - way classification model are :

- (i) Direct Method
- (ii) Short - Cut Method
- (iii) Coding Method

(I) Direct Method

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$ where $\mu_1, \mu_2, \dots, \mu_k$ are the arithmetic means of the k populations from which k samples are drawn at random.

The Alternative Hypothesis is $H_1 : \text{all } \mu_i (i = 1, 2, \dots, k) \text{ are not equal.}$

The steps in carrying out the analysis are given below.

(a) Calculation of Variance Between the Samples

It is the sum of the squares of the deviations of the means of various samples (or groups) from the grand mean. To calculate variance Between the Samples, we proceed, as follows :

(i) Calculate the sample means $\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_k$ of all the k samples.

(ii) Calculate the mean of the sample means or the grand mean $\bar{X}$ by using the formula

$$\bar{X} = \frac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3 + \dots + \bar{X}_k}{k} \text{ or } \bar{X} = \frac{T}{N},$$

where T = grand total of all observations and N = total no. of observations in all k

samples = $\sum_{i=1}^k n_i$ where n_i is the size of the i th sample.

(iii) Evaluate the deviations of the sample means from the grand mean i.e., find

$$\bar{X}_1 - \bar{X}, \bar{X}_2 - \bar{X}, \dots, \bar{X}_k - \bar{X}$$

(iv) SSB (or SSC) = Sum of the Squares of the variations between the samples
(or between the columns)

$$= \sum_{i=1}^k n_i (\bar{X}_i - \bar{X})^2$$

MSB (or MSC) = Variance or the Mean Square Between the samples
(or between the columns)

$$= \frac{SSB}{v_1}, \text{ where } v_1 = \text{degrees of freedom}$$

$$= (\text{no. of samples}) - 1 = k - 1$$

MSB indicates the degree of explained variance due to sampling fluctuations.

(b) Calculation of Variance Within the Samples

The variation within the samples would be on account of chance. This is because the same treatment is given to all items in the sample (the same fertilizer is being used on the sampled fields). The difference in the values of different items in a sample which is due to chance is called an estimate of the error. It is represented by SSE (Sum of squares due to errors).

Variance within the samples is calculated as shown below.

(i) Calculate the sample means $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_k$ of all the k samples.

(ii) Calculate the deviations of the various items of k samples from the mean values of the respective samples.

(iii) Square these deviations and obtain their total.

Thus SSW (or SSE) = Sum of the Squares of the variations within the samples
(or sum of the squares due to errors)

$$= \Sigma(X_1 - \bar{X}_1)^2 + \Sigma(X_2 - \bar{X}_2)^2 + \dots + \Sigma(X_k - \bar{X}_k)^2$$

MSW (or MSE) = Variance or Mean Square within the samples

$$= \frac{SSW}{v_2},$$

where $v_2 = \text{d.f.} = (\text{Total no. of observations}) - (\text{No. of samples})$
 $= N - k$

MSW indicates the chance variation and it cannot be explained in terms of variation in the population.

(c) Calculation of the Test Statistic F

Assuming that H_0 is true, the test statistic

$$F = \frac{MSB}{MSW} \text{ or } \frac{MSC}{MSE} = \frac{\text{variance between the samples}}{\text{variance within the samples}}$$

follows F - distribution with degrees of freedom $v_1 = k - 1$ and $v_2 = N - k$

In general $MSB > MSW$. But if $MSB < MSW$, we shall take the inverse ratio

i.e., $F = \frac{MSW}{MSB}$ i.e., variance ratio or F is the ratio between the greater variance and smaller variance.

(d) Decision

The computed value of F is compared with the critical (tabulated) value of F for $(k - 1, N - k)$ d.f. at $\alpha\%$ level (may be 5% or 1% level). If the calculated value $>$ the tabulated value, reject H_0 and conclude that all the proportion means are not equal. Otherwise, accept H_0 and conclude that the population means are all equal.

ANOVA TABLE (ONE-WAY CLASSIFICATION)				
Source of variation (SV)	Sum of squares (SS)	Degrees of Freedom	Mean Squares (MS)	Test statistic (F-ratio of variance)
Between Samples (Columns)	SSB	$k - 1$	$MSB = \frac{SSB}{k - 1}$	
Within Samples (Errors)	SSW	$N - k$	$MSW = \frac{SSW}{N - k}$	$F = \frac{MSB}{MSW}$
Total	SST	$N - 1$	—	—

$SST = \text{Total sum of squares of variations} = SSB + SSW$

In the above table SSB and SSW may be replaced by SSC and SSE, and MSB and MSW may be replaced by MSC and MSE.

8.22 ANOVA FOR TWO-WAY CLASSIFICATION (MANIFOLD CLASSIFICATION)

In two-way classification, observations (or data) are classified according to two different factors (or criteria). For example, fertilizers may be tried on different soil textures. The table dealing with data on such grouping has a number of columns and rows. Thus, with fertilizers in the column classification, the various types of land textures may be in rows. But there may be sampling variations besides the two factors considered which we call as 'residual variations'. The analysis in this case will get extended to include the sum of squares between rows, which was not there in one-way classification.

In this case the total variance or the sum of squares of variation consists of three parts :

- (i) SSC i.e., the Sum of Squares (or variance) between the columns (or due to columns)
- (ii) SSR, i.e., the Sum of Squares (or variance) between the rows (or due to rows) and
- (iii) SSE, i.e., the Sum of Squares for the residuals or residual variance (or the sum of squares due to errors). Thus

$$SST = SSC + SSR + SSE$$

If c be the number of columns and r the number of rows, then

The total number of degrees of freedom $= cr - 1$

Degrees of freedom between columns $= c - 1$

Degrees of freedom between rows $= r - 1$

Degrees of freedom between residuals $= (cr - 1) - (c - 1) - (r - 1)$

$$= (c - 1)(r - 1)$$

ANOVA TABLE for two-way classification is shown below :

ANOVA TABLE

Source of Variation (SV)	Sum of Squares (SS)	Degrees of Freedom (d.f.)	Mean Squares (MS)	Test statistic (Variance Ratio)
Between Columns	SSC	$c - 1$	$\frac{SSC}{c - 1} = MSC$	$F_C = \frac{MSC}{MSE} = \frac{\text{variance bet. columns}}{\text{Residual variance}}$
Between Rows	SSR	$r - 1$	$\frac{SSR}{r - 1} = MSR$	$F_R = \frac{MSR}{MSE}$
Residual (or Error)	SSE	$(c - 1)(r - 1)$	$\frac{SSE}{(c - 1)(r - 1)} = MSE$	e
Total	SST	$N - 1$		

Note that if $MSE > MSC$, then we shall take $F_C = \frac{MSE}{MSC}$

Similarly if $MSE > MSR$, then $F_R = \frac{MSR}{MSC}$

If the calculated value of $F_C >$ the table value of F at 5% level for $c - 1$ and $(c - 1)(r - 1)$ d.f., we reject H_0 and conclude that the difference between the column values is significant.

If the calculated value of $F_R >$ the table value of F at 5% level for $(r - 1)$ and $(c - 1)(r - 1)$ d.f., we reject H_0 at 5% level and conclude that the difference between the row values is significant.

Otherwise we accept H_0 .

Analysis of variance in two-way classification is illustrated below by an example.